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EDITORIAL

WILEY Terra Nova

HOW TO WRITE A GOOD ARTICLE FOR PUBLICATION IN *TERRA NOVA*

Although as Editors of *Terra Nova* we receive many well-written papers, occasionally some submissions are not as good as they could be in all aspects. Therefore, we have put together this guide on what we consider to be a well-written article for our journal. It may be particularly useful for younger and less experienced authors but even those who have written many papers may find some parts of this to be useful. However, please note that this is not a prescription for writing a paper acceptable to *Terra Nova*. If in your writing you have other ways of achieving the same objectives described below, we shall go on welcoming the diverse range of submissions to *Terra Nova*.

HOW TO WRITE A GOOD ARTICLE FOR PUBLICATION IN *TERRA NOVA*

1 | Defining a single, main result

Because *Terra Nova* specializes in publishing SHORT, innovative and/or provocative papers, the most important rule for publishing in our journal is to focus the entire manuscript on a SINGLE FINDING (or result).

Your first task is therefore to DEFINE this finding in a concise statement (sentence). Note that this statement is critical as it will be used at every step of the writing of your manuscript. Take the time to write this statement as concisely and precisely as possible, in such a way that it captures the essence but also the entire content of your finding. It is also essential that all co-authors contribute to and/or agree with this statement. This will greatly help the process of jointly writing a well-focused and coherent manuscript.

The next essential ingredient is to convince yourself that this finding is NOVEL (i.e., it has not been published in another journal/article) and is TIMELY (i.e., it is of interest to the broad Earth Sciences community and *Terra Nova* readership). For this you must have a good knowledge of the field and of the main issues currently debated in this field. You must also be able to convince yourself that your result contributes to these issues in a meaningful, i.e., non-negligible, manner.

2 | Preparing a first draft

A *Terra Nova* article typically contains the following sections/parts:

- an Abstract,
- a list of Authors and Affiliations,
- a set of Keywords,
- an Introduction,
- a Methods section,
- a Results section,
- a Discussion,
- a Conclusions section,
- the Acknowledgements,
- a list of References.

In some cases, other sections may be envisaged, such as a “Geological setting” section, for example. The key to a successful manuscript is to be aware of the CONTENT of each of the sections and HOW THEY RELATE TO EACH OTHER (see Figure 1). In the following, we will go through the various parts/sections in the order in which they will appear in the manuscript. Note, however, that this is NOT the order in which they should be written. A good plan is to start by writing (a) the Results section, followed by (b) the Methods section, (c) the Introduction, (d) the Discussion and (e) the Conclusions section (see Figure 1). The Abstract, the Title and the Keywords are the LAST THING you will write. We also strongly advise you to write a first draft of your manuscript as a series of statements or bullet points under each of the section headings given above that will later form the substance/idea for each of the paragraphs of your manuscript.

3 | Structure of a good manuscript

No matter what excellent points a paper may make, these points will be worthless if nobody reads your work. It is vitally important that the title and the beginning of the Abstract persuade the reader to read on.

The **TITLE** is the first point of contact with the reader. It should inform the reader about the content of your manuscript. It is also used by search engines to find *your* paper among thousands of other papers related to a similar topic. It is therefore extremely important that it be short, yet informative (i.e., it describes your finding and places it in its right context). Ideally, it should contain as many as possible of your keywords (see below), and it should be searchable. For example, using pop-culture terms that will attract many spurious search-hits could backfire.

The **LIST OF AUTHORS** (and their AFFILIATIONS) should include all persons/researchers having made a significant contribution to the

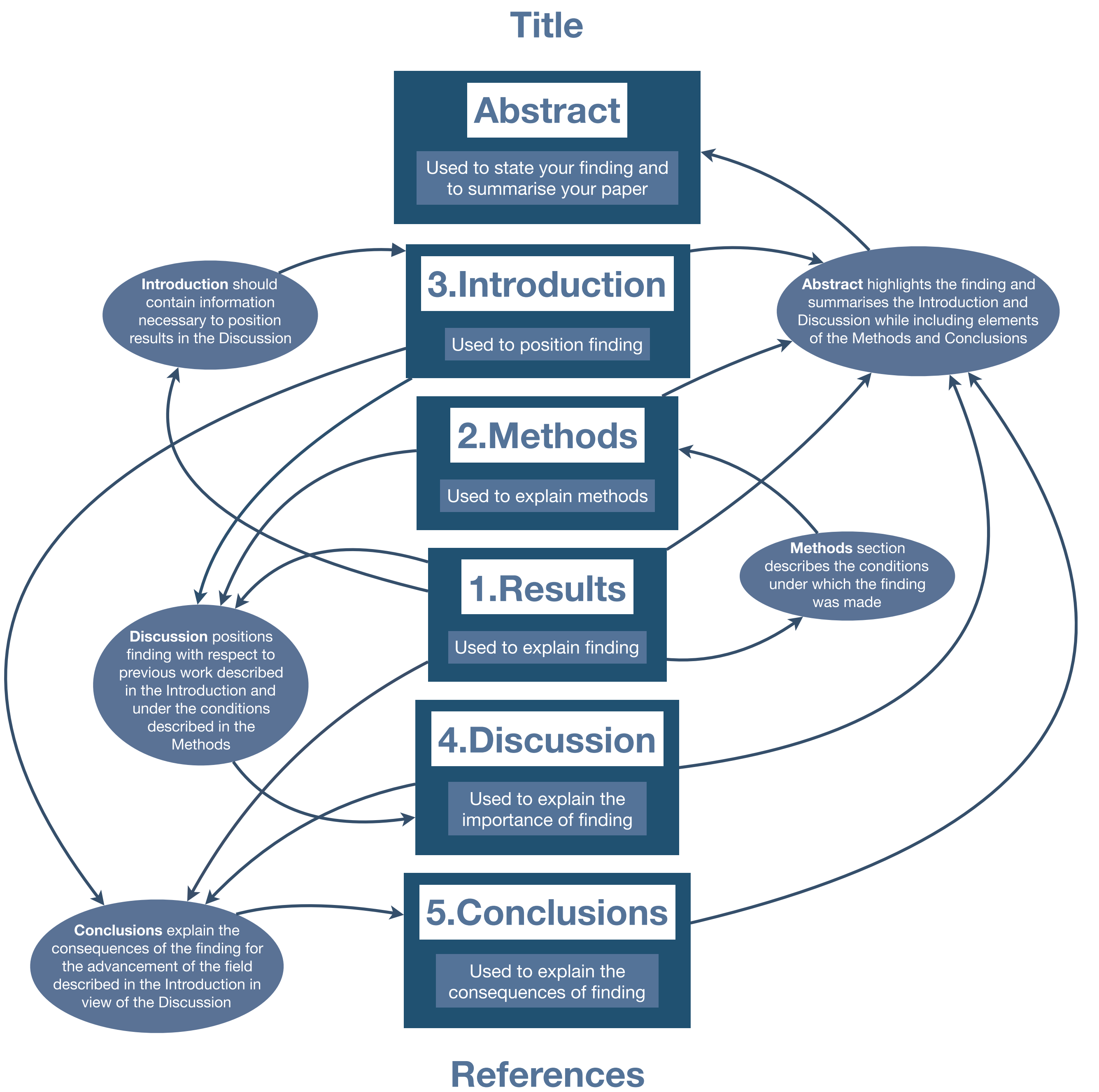

FIGURE 1 Flow chart explaining how the different elements of your manuscript interact with each other to optimize the flow of information to your reader and readability

work presented in your manuscript and to the preparation of the manuscript, and ONLY those.

The **ABSTRACT** starts the process of why anyone should bother to read your paper. It is, by definition, a short yet complete summary of your paper. *Terra Nova* is a journal covering the widest range of geoscience topics and is read by a very broad range of scientists. To get the audience of many different specialists to read beyond the first sentence of the abstract, that first sentence must engage each of them and provide a hook to draw them into reading further. The easiest way to do this is to state in the first sentence of the abstract the generic scientific problem addressed by the paper and that you have now solved it. This sentence should be understood by a scientist in any discipline. The abstract must also contain a sentence that describes your main finding. That sentence should start with "We have found that…" or "We show that…" or "We demonstrate that…". It should also contain elements describing the context of your work, including key hypotheses you have made or methods you have used, as well as a sentence comparing your work to existing knowledge/work and explaining the importance/relevance of your work. The abstract is one of the most important sections of your manuscript as it will be used by the editor to assess whether it should be sent out for review and to select reviewers. Later, when your paper is published, it is one of the key elements (with the title and the keywords) used by search engines to find and choose your paper.

The **KEYWORDS** are used (mostly by search engines) to LOCATE your work/paper in vast knowledge repositories. If you want your paper to be read and cited, you must use keywords that will be used by the researchers who are most likely to be interested by your finding. A good start is to go through your manuscript (once it is completed) and find the words/terms that are used many times in the text. Add more general terms, i.e., terms that describe the broad field/subject of your finding, and more specific terms, i.e., the location of your study, a specific method you have used, etc. Add abbreviations that may be relevant too. The best way to verify the quality of your keywords is to type them in a search engine and check that the results match the subject of your paper.

The **INTRODUCTION** section contains three essential parts. The first part is a statement about the topic of the manuscript and WHY IT IS IMPORTANT to the community, to the advancement of knowledge, to society, etc. This part of the Introduction is usually written in the present tense and will be an expansion of the first sentence of the Abstract. The second part contains a concise, yet complete summary of the state-of-knowledge or STATE-OF-THE-ART on the subject of your finding. You must cite the papers that initiated the main debate(s) or posed the main questions you are trying to address, as well as all the main contributions made so far to address these questions. Beware of not limiting yourself to your own papers on the subject or to those of your close collaborators, and to give proper credit to ALL opinions/ideas published on the subject. The purpose of this part of the Introduction is to provide the elements that will allow you, in the Discussion section of your manuscript, to position your finding (and demonstrate that it is novel, important, worth publishing,…) with respect to what is known and what has been done/published before. You should therefore include ALL relevant literature (i.e., that you will cite again in your Discussion section), but ONLY the relevant literature. Carefully consider the need to include references that you will not use/cite in your Discussion section. The second part of the Introduction is commonly written in the past tense. The third part of the Introduction contains the outstanding questions that need to be addressed on the topic of your paper, the essential data or constraints that need to be collected/assembled. When writing this part of the Introduction, you should make sure that your finding does address one or several of these questions and/or contributes to what you describe as essential to the advancement of knowledge. The content of the Introduction will be used by the editor to judge the relevance and appropriateness of the subject of your manuscript for publication in *Terra Nova*.

The **METHODS** section provides all the elements that are necessary to understand what you have done to achieve the results that form the basis of the paper. IT SHOULD GIVE ALL NECESSARY ELEMENTS/INDICATIONS TO REPRODUCE YOUR RESULTS. It should also explain why the method(s) you have chosen is(are) relevant to address the big question at the centre of your paper. You do not need to explain every method used in great detail if they have already been published elsewhere. However, if you have used a new/different/modified method, you must explain it. This part of the manuscript is usually easy to write as it is technical and authors must be, by definition, experts in the techniques they have used.

The **RESULTS** section contains YOUR MAIN FINDING and, if necessary, intermediary results that lead to the main finding. These include a description of the observations you have made and of the data you have produced/collected, including uncertainties and/or error estimates. Usually the Results section is short. You should clearly differentiate between an observation and its interpretation. It is recommended to start writing the Results section (and the manuscript) by preparing the figure(s) that describe/report your main finding and any important intermediary result. The Results section becomes a simple narrative based around the description of the figure(s). See the paragraph on FIGURES below for advice on preparing a figure.

The **DISCUSSION** section is the most critical part of your manuscript. This is where you have to convince your readership (and the editor and the reviewers) that your finding is novel and relevant. For that you have to compare and contrast your finding to previous work on the subject and clearly state in which way(s) it improves, complements, contradicts or, at minima, confirms previous findings. The Discussion should also discuss the limitations of your findings, as well as remind the reader of the hypotheses under which you have worked and made your finding, thereby clearly defining the boundaries of applicability of your work. This is the section that the editor reads most carefully before deciding whether to send the manuscript to reviewers. This is therefore where you convince the editor that your finding is innovative and/or provocative.

The **CONCLUSIONS** section should start with a sentence summarizing your finding, followed by statements about the "new" state-of-knowledge of the field, i.e., following your finding, the consequences of your findings for the field, as well as indications or suggestions for future work.

The **ACKNOWLEDGEMENTS** should contain a list of contributors who cannot be considered as authors of the manuscript. Financial support and/or sources of conflict of interest should also be acknowledged.

The **REFERENCES** section should contain a list of all references cited in the text. The format used by *Terra Nova* is explained in the Author Guidelines. It is good to remember that citations must clearly relate to a point made in the manuscript and that they should be located just after that point is made.

4 | Writing the manuscript

Once you have written the skeleton of your manuscript in bullet points, and agreed with all your co-authors on the content of each section, you may start writing the manuscript in its final form. For this, you must transform each bullet point into a **PARAGRAPH**. Remember that a paragraph must correspond to an idea/concept that you wish to develop. A paragraph therefore contains at a minimum

THREE SENTENCES: an opening sentence, a body made of one or several sentences, and an end sentence. In general, the readability of your manuscript will be greatly enhanced if the final sentence of a paragraph provides a link to the next one.

A **SENTENCE** is a step in a logical unit; it must contain one verb, one subject and one object. Avoid long sentences. Sentences are terminated by full stops and can contain commas to help the flow of the sentence. If a sentence contains more than four commas, it should be split into smaller sentences. In writing your sentences, remember that you should choose WORDS for their clarity and precision. If you are not confident about your knowledge of the English language, you must ask a competent English speaker to proof-read and/or edit your manuscript before submitting it to *Terra Nova*. A link to an English Language Editing service is also provided on *Terra Nova*'s web page.

5 | Preparing figures and tables

A **FIGURE** works best when it is used to help you explain ONE concept, ONE idea, ONE method or ONE result. This is especially true when you submit your work to *Terra Nova* where there is no limit on the number of figures. The main purpose of a figure is to facilitate comprehension and improve accessibility, not the opposite. It must therefore be prepared with great care. A good figure has a single message and that message must be transmitted to the reader rapidly/efficiently, i.e., almost at first sight. When preparing a figure, you must start by writing down the message that the figure is meant to transmit to the reader and decide on the most efficient way (the type of plot, diagram, how many panels in the figure, etc.) to use for that transmission.

Consider a figure as a structure made up of elements (axes, curves, dots, map, picture, etc.) that must be chosen and organized to highlight the message. For example, if you plot two or more curves in a simple x–y diagram, it is because they need to be compared to each other, not because you want to save space. If you need to locate something on a map, make sure that that element appears in a bright, easily differentiable colour. Make sure that the fonts you use for labels are easily readable (size and type), and that line thicknesses are adequate and proportional to the importance of the elements they represent.

Figure captions should contain a short descriptor of what is shown in the figure as well as elements that are essential to understand the figure but could not be directly included in the figure for clarity. This includes long names or explanatory labels, references to data shown in the figure, etc. If the figure is based on a previously published figure, you should say so in the caption (modified from…, or reproduced with permission from…). Note that all figures must be referred to in the text/body of the manuscript. "Orphan" figures (i.e., those that are not referred to in the text) are not allowed. The numbering of the figures should follow the order in which they are cited in the text.

A **TABLE** is used to report a large collection of numbers such as data/measurements or model parameter values to facilitate their comparison and/or to clarify their presentation. For example, each measurement that you made on a sample should appear on a single line of a table with the name/label of the sample given in the first column. No quantitative result should be presented with more significant figures/decimal digits than the statistically valid precision/uncertainty of the measurement, and the precision should be stated in the caption or text. Note that in many fields of investigation in the Earth Sciences, there is a standard way to present data in a table. You should refer to previously published papers that use a similar technique/method and follow the same structure. Tables have a caption and, contrary to figures, a short title usually placed on top of the table.

6 | To *Terra Nova*, and beyond

We hope this synopsis is of use to authors still working to master the writing of a short scientific paper. A well-defined structure typically makes it easier to write an effective paper, freeing your attention from thinking about the structure of your paper to focusing on communicating the contents of your finding. Once you are comfortable with writing papers in this structure, you may decide at times that this structure could be changed for even more effective communication of a particular finding. We wish you productive writing!

Jean Braun
Max Coleman
Jason Phipps Morgan
Carlo Doglioni
Georges Calas
Klaus Mezger
Scientific Editors, Terra Nova